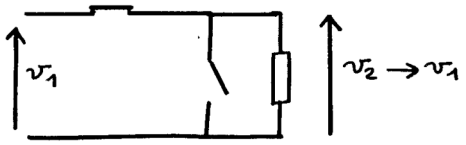
Corrigé du DS-commun (Partie electronique)

MP-1/CPGE-Laâyoune

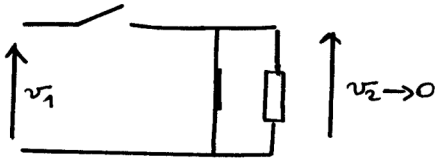
Etude d'une cellule de filtrage

1) Etude qualitative :

- en basse fréquence $L\omega \to 0$, $\frac{1}{C\omega} \to \infty$



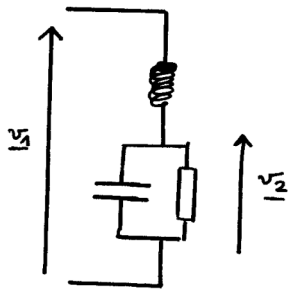
- en haute fréquence $L\omega \to \infty$, $\frac{1}{C\omega} \to 0$



Le filtre est donc un **passe-bas**

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2) Fonction de transfert



Par les diviseurs de tension :

$$\underline{H} = \frac{\underline{v_2}}{\underline{v_1}} = \frac{\underline{Z}_{R//C}}{\underline{Z}_{R//C} + \underline{Z}_L} = \frac{1}{1 + \underline{Z}_L \underline{Y}_{R//C}} = \frac{1}{1 + jL\omega\left(\frac{1}{R} + jC\omega\right)}$$

$$= \frac{1}{1 + j\frac{L\omega}{R} - LC\omega^2}$$

$$= \frac{1}{1 + \frac{j}{Q}\frac{\omega}{\omega_0} - \frac{\omega^2}{\omega_0^2}}$$

$$\boxed{\underline{H} = \frac{1}{(1 - x^2) + j\frac{x}{Q}}}$$

(remarque : par Millman, on obtient

$$\underline{v_2} = \frac{\underline{v_1}\,\frac{1}{jL\omega}}{\frac{1}{jL\omega} + jC\omega + \frac{1}{R}}$$)

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3) $$G = \frac{1}{\sqrt{(1-x^2)^2 + \frac{x^2}{Q^2}}}$$

L'extremum de G, c'est aussi l'extremum de

$$D_{(u)} = (1-u)^2 + \frac{u}{Q^2} \qquad (\text{avec } u = x^2)$$

$$\frac{dD}{du} = -2(1-u) + \frac{1}{Q^2}$$

Il y a extremum pour :

$$2(1-u) = \frac{1}{Q^2}$$

$$u = 1 - \frac{1}{2Q^2}$$

($u \to x^2$)

Il faut donc que $1 - \frac{1}{2Q^2} > 0$

soit $$\boxed{Q > \frac{1}{\sqrt{2}}}$$

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4) A l'extremum, on a donc :

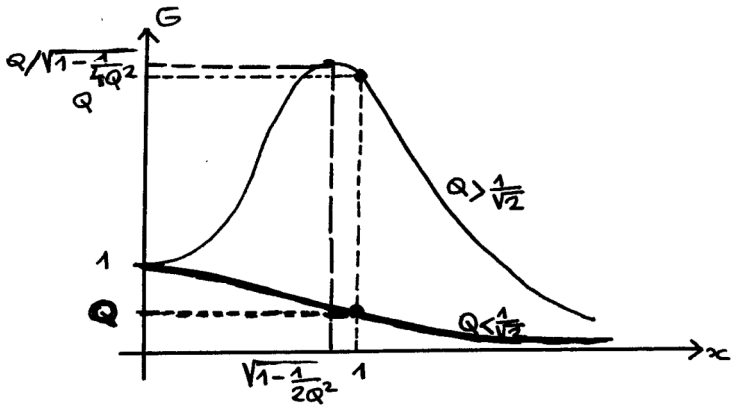
$$\boxed{\omega_r = \omega_0 \sqrt{1 - \frac{1}{2Q^2}}}$$

et : $$G_{max} = \frac{1}{\sqrt{\left(\frac{1}{2Q^2}\right)^2 + \frac{1}{2Q^2}\left(1 - \frac{1}{2Q^2}\right)}}$$

$$G_{max} = \frac{Q}{\sqrt{1 - \frac{1}{4Q^2}}}$$

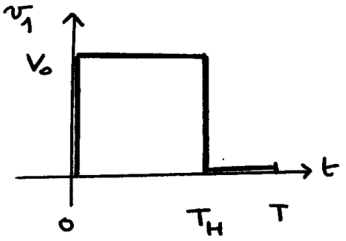
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5)



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6)



$$V_{moy} = \frac{1}{T}\int_0^T v_1(t)\,dt$$

$$= \frac{1}{T}\left(\int_0^{T_H} V_0\,dt + \int_{T_H}^{T} 0\,dt\right)$$

$$= V_0\,\frac{T_H}{T}$$

$$\boxed{V_{moy} = V_0\,\alpha}$$

---

7)

$$\boxed{\omega_{fond} = \omega}$$

$$V_{fond} = \frac{V_0}{\pi}\left(\underbrace{\sin^2(2\pi\alpha)}_{4\sin^2(\pi\alpha)\cos^2(\pi\alpha)} + 4\sin^4(\pi\alpha)\right)^{1/2}$$

$$\boxed{V_{fond} = \frac{2V_0}{\pi}\sin(\pi\alpha)}$$

$$\boxed{\omega_{harm} = 2\omega}$$

$$V_{harm} = \frac{V_0}{2\pi}\left(\sin^2(4\pi\alpha) + 4\sin^4(2\pi\alpha)\right)^{1/2}$$

$$\boxed{V_{harm} = \frac{V_0}{\pi}\,|\sin(2\pi\alpha)|}$$

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8) Pour le continu, on utilise la fonction de transfert en continu

$$V_{2,m} = \underbrace{G(\omega=0)}_{=1} V_{1,m}$$

$$\boxed{V_{2,m} = V_{1,m} = V_0\alpha}$$

---

9) A.N.

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$\boxed{f_0 = 112{,}5\ \text{Hz}}$$

$$Q = \frac{R}{L\omega_0} = R\sqrt{\frac{C}{L}}$$

$$\boxed{Q = \frac{1}{\sqrt{2}} = 0{,}707}$$

$$V_{2,m} = V_0\,\alpha$$

$$\boxed{V_{2,m} = 5\ \text{V}}$$

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10) On a $f\ (5\text{kHz}) \gg f_0\ (0{,}113\ \text{kHz})$

Donc pour les deux harmoniques (f et 2f) on pourra faire $G_{(f)} = \dfrac{1}{\sqrt{(1-x^2)^2 + \frac{x^2}{Q^2}}} \simeq \dfrac{1}{x^2}$ ou $\left(\dfrac{f_0}{f}\right)^2$

⟶ harmonique 1 (f)

$$V_{fond,\,sortie} = \left(\frac{f_0}{f}\right)^2 V_{fond}$$

$$= \left(\frac{f_0}{f}\right)^2 \frac{2V_0}{\pi}\sin(\pi\alpha)$$

$$\boxed{V_{fond,\,sortie} = 4{,}19\ 10^{-3}\ \text{V}}$$

⟶ harmonique 2 (2f)

$$V_{harm,sortie} = \frac{f_0^2}{4f^2} V_{harm}$$

$$= \frac{f_0^2}{4f^2} \frac{V_0}{\pi} |\sin(2\pi\alpha)|$$

$$\boxed{V_{harm,sortie} = 0{,}52\ 10^{-3}\ V}$$

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11) • Pour $V_{2,alt}$ on prendra $\underline{4{,}19\ 10^{-3}\ V}$

• (cf on néglige les autres harmoniques : $0{,}52 \ll 4{,}19$)

• taux d'ondulation

$$\frac{\Delta V_2/2}{V_{2,moy}} = \frac{\left(\frac{f_0}{f}\right)^2 2 \frac{V_0}{\pi} \sin(\pi\alpha)}{\alpha V_0}$$

$$\boxed{\tau_{ond} = \left(\frac{f_0}{f}\right)^2 2 \sin(\pi\alpha)}$$

$$= 0{,}08\ \%$$
